NAME ________________________ DATE ____________ PERIOD ________

Paper Practice 9.5

Geometric Sequences

WORKED KEY (odd-numbered items) – Show all of your work in the space provided. Use $a_n = a_1 \cdot r^{(n-1)}$ where a formula is needed. Round money to the nearest whole unit.

1. Determine whether each sequence is geometric. If it is, state the common ratio. (Is It Geometric?)

a. −432, 144, −48, 16, … ANSWER: Geometric, r = −⅓

b. 4, 9, 25, 36, … ANSWER: Not geometric – 9/4 ≠ 25/9.

c. 16, 12, 8, 4, … ANSWER: Not geometric – the terms drop by a constant 4, so it is arithmetic.

2. Find the next three terms in each geometric sequence. (Next Terms)

a. 64, 16, 4, 1, …

b. 8, 12, 18, 27, …

c. 3, −6, 12, −24, …

3. Find the next three terms in each geometric sequence. (Next Terms)

a. 96, 48, 24, 12, … ANSWER: r = ½: 6, 3, 1.5

b. 2, 10, 50, 250, … ANSWER: r = 5: 1250, 6250, 31250

4. Write an explicit formula $a_n = a_1 \cdot r^{(n-1)}$ for each sequence. (Explicit Formula)

a. 5, 15, 45, 135, …

b. 800, 400, 200, 100, …

5. A sequence has $a_1 = 6$ and r = −2. Write the first five terms. (Build a Sequence) ANSWER: 6, −12, 24, −48, 96

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6. Use an explicit formula to find the indicated term. (Find a Term)

a. 512, 256, 128, 64, … find a_{11}

b. −8, 24, −72, 216, … find a_9

7. Use an explicit formula to find the indicated term. (Find a Term)

a. 5, 3, 1.8, 1.08, … find a_8 ANSWER: $r = 0.6$; $a_8 = 5(0.6)^7 = 0.139968$

b. 7, 21, 63, … find a_6 ANSWER: $r = 3$; $a_6 = 7(3)^5 = 7(243) = 1701$

8. North Dakota's population was 685,242 in 2011 and has been increasing about 2.5% each year. Estimate the population in 2030. (Apply)

9. Global vinyl record sales were $267 million in 2014 and have been increasing about 26% each year. Estimate sales for 2025. (Apply) ANSWER: 11 years: $267(1.26)^{11} \approx \$3{,}393$ million (about $3.4 billion)

10. Error Analysis – Gus says 2, 4, 6, 8, … is geometric because the terms keep increasing. Identify his mistake. (Error Analysis)

11. Explain the difference between an arithmetic and a geometric sequence, and give an example of each. (Explain) ANSWER: Arithmetic adds a common difference; geometric multiplies by a common ratio. Answers vary.